

CP MAS Kinetics Study of Ionic Liquids Confined in Mesoporous Silica: Convergence of Non-Classical and Classical Spin Coupling Models

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Abstract The high data point density measurements of $^1\text{H} \rightarrow ^{11}\text{B}$ cross-polarization (CP) kinetics upon magic-angle spinning (MAS) of [bmim][BF₄] confined in mesoporous SBA-15 and MCM-41 were carried out. The complex shaped ^{11}B CP MAS signals were observed in both silica and decomposed into two Lorentz components. This points towards the possibility of bimodal distribution of [bmim][BF₄] in the studied confinements. The convergence of classical and non-classical spin coupling models was deduced processing CP kinetic curves. A good fit of the theoretical curves to the experimental data was achieved using both models without any non-random deviations between theory and experiment to appear. The convergence of spin coupling models was discussed in terms of relatively high mobility of BF₄[−] anion respect to the cation and the dynamics of anions in pores. These factors delete the borders between spin clusters. The spin diffusion along the pore surfaces in MCM-41 is more than twice faster than in SBA-15.

1 Introduction

Cross-polarization (CP) method is widely used in solid-state NMR studies [1, 2]. Usually CP is applied combined with magic-angle spinning (MAS) technique for the enhancement of signals from dilute nuclei by the polarization transfer from abundant ones, typically protons. As CP is based on the heteronuclear dipolar interactions, it is intrinsically sensitive to internuclear distances and to dynamic displacements of molecules or functional groups therein. Hence, CP looks to be a powerful tool to determine the spatial distribution of coupled nuclei as well as to

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monitor the molecular mobility in solids via the spin polarization transfer rates [3]. Therefore, despite the signals may even be weaker than that those obtained from a conventional pulse-acquire Bloch-decay (BD) experiment [4], CP experiments with the polarization transfer between abundant nuclei are carried out more and more often [5–14].

The CP MAS technique was applied in the present work for the study of ionic liquid 1-butyl-3-methyl-imidazolium tetrafluoroborate ([bmim][BF₄]), confined as a guest compound in mesoporous silica (SBA-15 and MCM-41). This choice was due to an unabated interest to room temperature ionic liquids (RTILs). This interest is caused by the wide areas of application of RTILs as neat compounds as well as the additives to multifunctional composites in advanced industry, separation, purification, catalysis, and other technological processes [15–18]. The class of RTILs can be considered as one of the most successful breakthroughs in the field of ‘smart’ materials and multifunctional compositions. Materials are classified as being ‘smart’, if they possess an intrinsic ability to sense and definitely to respond in a predictable way to various external stimuli (see [19] and the Refs therein).

It is known that mesostructured silica of MCM-41 and SBA-15 types constitute two-dimensionally hexagonal arrays of cylindrical pores of uniform sizes that cover the range 2–10 nm [20]. Because of the high density of pores and the relatively small pore diameters, these silica materials have extremely large inner surfaces, relative to the size of the individual particle. Moreover, the geometry of the pores is highly anisotropic (a preferred axis exists in the direction of the pore cylinder axis), and the diffusion of guest molecules in the pores is found to exhibit deviations from ordinary diffusion [21]. The research of RTILs in the combination with such materials is very intriguing and challenging. Any discovered unexpected feature of such composites can suddenly open new horizons creating novel generation of sensors, controllers, molecular machines, etc.

CP MAS kinetics, i.e., the dependence of the NMR signal intensity on the contact time for interacting spins, has reveal its capability in resolving fine structural surface and bulk effects as well as dynamical details in complex solids [13, 14]. The purposes of the present work were: (i) high data point density measurements (128 equidistant points over 0–0.01 s contact time range) of ¹H→¹¹B cross-polarization kinetics upon MAS of [bmim][BF₄] in SBA-15 and MCM-41; (ii) to prove the CP MAS as an experimental tool to reveal fine structural details in such novel composites as well as to evaluate the capability of existing kinetic models to describe CP kinetics therein; and (iii) to obtain novel information concerning the physical origin of CP transfer regimes and spin coupling in ionic liquids confined in mesostructured porous materials.

2 Materials and Methods

Mesostructured SBA-15 type amorphous silica and MCM-41 type aluminosilicate were obtained from Sigma-Aldrich. Both these materials are characterized having the hexagonal structure of cylindrical pores. The pore diameters are 3.3 nm and 7.9–8.0 nm, and the specific pore volumes are 0.93 cm³g^{−1} and 0.96–1.01 cm³g^{−1}

for MCM-41 and SBA-15, respectively. To minimize the amount of water, both silica after packing in MAS rotors were dried in vacuum at 110 °C for 24 h. Immediately, after drying the ionic liquid 1-butyl-3-methyl-imidazolium tetrafluoroborate ([bmim][BF₄]), also from Sigma-Aldrich and used without further purification, was added to both samples. The weight ratios for SBA-15/[bmim][BF₄] and MCM-41/[bmim][BF₄] samples were 33.1/6.4 mg and 28.7/14.4 mg, respectively. Taking these mass ratios as well as the MCM and SBA specific pore volumes and the surface area data, the pores fillings of ca 45 and 16% were determined for the MCM and SBA samples, respectively.

All NMR measurements were carried out using Bruker AVANCE III HD 400 NMR spectrometer with the 4 mm CP MAS probe head at 25 °C. The resonance frequencies were 400 MHz and 128 MHz for ¹H and ¹¹B nuclei, respectively. In all measurements, the MAS frequency was 5 kHz. The Hartmann–Hahn (HH) matching condition $\omega_H - \omega_S = n \omega_{MAS}$ was fulfilled for the sideband order $n = -1$. The B_1 field values (nutation frequencies) for CP were 63,500 Hz and 68,500 Hz for ¹H and ¹¹B, respectively. ¹H→¹¹B CP kinetics was measured using a rectangular contact pulse. The pulse duration was varied between 50 μs and 10 ms with the increment of 78 μs. For each increment, 256 scans were accumulated with the repetition delay of 2 s. The obtained data were processed using *Topspin 3.2* program and the signal contour separation was performed using *Dynamics Center* software. The further data analysis was carried out using *Microcal Origin Pro 9* package.

3 Results and Discussions

The I→S CP technique, where both I and S are natural abundant nuclei, is useful for the characterization of complex solid materials. The conventional Bloch-decay (BD) NMR experiment provides information on S nuclei in all possible manifolds, while the ¹H→S CP NMR (i.e., I = ¹H) favours those located close to protons. This creates a certain segregation of S nuclei and takes into the scope of the running experiment only some of them. Moreover, this opens the possibility to manipulate with the spatial scale of such segregation by varying the CP contact time and sequent processing of the so-called CP kinetic curves. It provides a chance; however, it does not vouch that certain fine structural or dynamic details that are hidden in BD experiments will be revealed in CP ones.

CP MAS kinetics study of RTIL confined in mesostructured materials, as far as we know, has been carried out for the first time. Namely, ¹H→¹¹B CP MAS kinetics technique was applied for the study of [bmim][BF₄] as a guest compound in mesoporous silica of MCM-41 and SBA-15 types. This choice of interacting nuclei pairs provides the possibility to follow the evolution of the communication between spins in the anion (¹¹BF₄) and the cation (all ¹H are in bmim⁺ and no protons in the anion). Thus, ¹H→¹¹B CP MAS kinetics contains the information about the mutual dynamics of both ionic parts in RTIL and the structural changes therein. Note that the ionic liquid [bmim][BF₄] is among the most commonly investigated RTILs [22]. It was chosen also for the present study because of its air/moisture tolerance, low

melting point, viscosity, and surface tension [23]. Thus, the good penetration into the pores of MCM-41 and SBA-15 can be expected. Besides, the signal with a partially resolved feature of very weak ^{19}F – ^{11}B coupling is seen in the ^{11}B NMR spectrum of the neat [bmim][BF₄] (Fig. 1). The coupling constant $^1J(^{19}\text{F}$ – $^{11}\text{B}) = 1.2$ Hz was determined by the processing of the line shape. Thus, it can be expected that the shape of ^{11}B signal might be very effective tool to sense any change in spin sites due to confinement.

The results of ^1H → ^{11}B CP MAS experiments look promising indeed. The complex shape of the ^{11}B CP MAS signals was observed in MCM-41 and SBA-15 (Fig. 1). The lines were decomposed into two Lorentz components denoted as ‘narrow’ and ‘broad’. This points to the possibility of bimodal distribution of [bmim][BF₄] in mesostructured silica, similarly, as it was observed in the case of water as a guest molecule [20]. The CP kinetics data were processed using two spin coupling models for each kinetic curve separately.

3.1 Theoretical Models

There are two spin coupling models that are most often applied in the processing of CP kinetic data: the classical model, denoted as I–S (in the present work, the spins I and S are protons and ^{11}B , respectively) and the non-classical one, I–I*–S, where the asterisk denotes spins I being in close neighborhood to S spins [3]. In the first case, it is assumed that spin diffusion is fast enough to force spins to behave as a system with a uniform spin temperature [3, 24]. Starting from the rigorous (however, rather complex) formula that describes the CP dynamics between two nuclei [24], one can simplify the dependence of the CP signal intensity $I(t)$ on the contact time t reducing it to the bi-exponential decay function, if S nuclei are highly diluted among I nuclei, i.e., $N_S \ll N_I$, and the CP process is fast enough comparing with the spin–lattice relaxation of spin S in the rotating frame:

$$I(t) = I_0(1 - T_{\text{IS}}/T_{1\rho})^{-1} \left[e^{-\frac{t}{T_{1\rho}}} - e^{-\frac{t}{T_{\text{IS}}}} \right], \quad (1)$$

where $1/T_{\text{IS}}$ is the CP rate constant and $1/T_{1\rho}$ is the spin–lattice relaxation rate of spin I in the rotating frame.

In the case of the I–I*–S model, the exchange of polarization in I*–S spin pairs or even in larger spin clusters $\text{S} - \text{I}_N^*$ runs in the oscillatory manner mimicking the energy exchange between a pair of coupled pendulums [3]. The CP kinetics exhibiting oscillations of intensity in the short contact time range are often observed for a broad set of materials, those as highly regular and disordered crystals as well as nano-structured compounds [3–6, 8, 12–14, 25–29]. Already in the pioneer works [25, 26], there was deduced that the frequency of oscillation depends on the orientation of a crystal in the external magnetic field and it is equal to one-half of the dipolar splitting b , which depends on the distance r between two interacting nuclei (I and S) and on the angle θ between the $\mathbf{r}$ vector and the magnetic field direction:

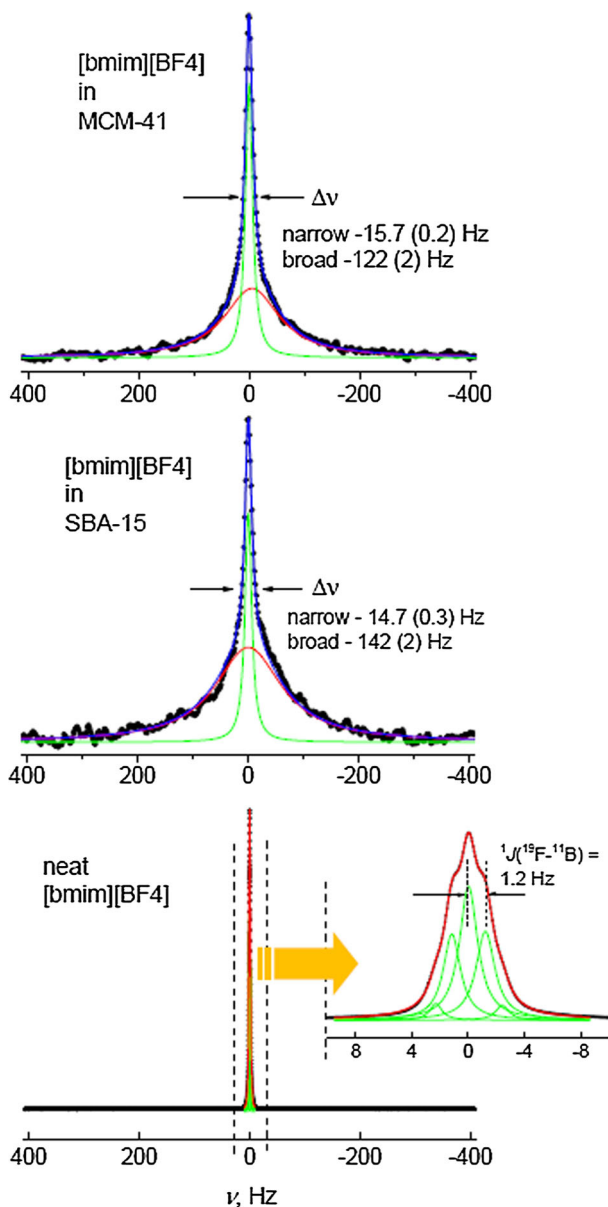


Fig. 1 ^{11}B CP MAS line shapes of [bmim][BF₄] as a guest compound in MCM-41 and SBA-15, the contact time is 3 ms (two upper panels). The errors determining the line widths are given in the brackets. ^{11}B NMR signal in the neat [bmim][BF₄] is shown for comparison (the lower panel). The one-bond ^{19}F - ^{11}B coupling was detected and the coupling constant is determined (1.2 Hz). All measurements were carried out at 25 °C

$$b = \frac{\mu_0 \gamma_I \gamma_S \hbar}{4\pi r^3} \frac{(1 - 3 \cos^2 \theta)}{2} = D_{\text{IS}} \frac{(1 - 3 \cos^2 \theta)}{2}. \quad (2)$$

The oscillations get damped by the subsequent spin diffusion, as well as they can be blurred at the extended spectral disorder in the spin system [13]. In other words, the distribution of b values (b -spectrum) is broad enough. The later factor can be characterized by the time constant T_2 that should be approximately equal to the spin–spin relaxation time at a very long contact time [13, 14]. Then, the CP kinetic curve $I(t)$ can be fitted by

$$I(t) = I_0 e^{-t/T_{1\rho}} \left[1 - \lambda e^{-\frac{t}{T_{\text{dif}}}} - (1 - \lambda) e^{-\frac{3t}{2T_{\text{dif}}}} e^{-\frac{t^2}{2T_2^2}} \right], \quad (3)$$

where T_{dif} is the spin-diffusion time constant and λ is the parameter, which describes the distribution of polarization in the S– I_N spin cluster at the initial stage of CP. The parameter λ depends on the cluster composition as well as on the molecular motion [3, 4, 7]. There was also claimed that for a rigid lattice, $\lambda = 1/(N + 1)$. However, this parameter has to be adjusted for each system during the fitting of the model curve to the experimental data set.

Equation 3 is the most often used one for the processing of experimental CP kinetic data in the cases when it was deduced that the non-classical I–I*–S model was more appropriate, comparing to the classical I–S model [3, 4, 7, 29]. However, it is sometimes not a simple task to recognize the model appropriateness or its validity. As it was noted in Ref. [3], to discriminate the different CP regimes, the CP kinetic curves have to be sampled very precisely, particularly, in the short contact range. Indeed, an interesting example on this problem is given in Ref. [29], where the validity of CP models for nano-structured calcium hydroxyapatite and that containing amorphous phosphate phase (ACP–CaHA) was analyzed. In the case of ACP–CaHA, the nice coincidence with experiment (the correlation coefficient $R^2 = 0.993 \div 0.999$) was achieved over the wide contact time range (0–10 ms) and quite realistic values of the fit parameters have been obtained using both models (I–S and I–I*–S). However, if the classical I–S model was applied, some non-random discrepancies between theory and experiment were yet noticed [29]. Those systematic discrepancies were possible to reveal only thanks to very high experimental point density (500 points over the whole contact time range of 10 μs –10 ms at the sampling frequency of $5 \times 10^4 \text{ s}^{-1}$). Only then, it was possible to conclude that the non-classical I–I*–S spin coupling model was more preferable to describe the CP kinetics in the studied system.

To avoid such incertitude concerning the validity of the models in the present work, we have started from the equation derived in Ref. [25] for the isolated spin pair, namely:

$$I(t) = I_0 e^{-\frac{t}{T_{1\rho}}} \left[1 - \frac{1}{2} e^{-k_2 t} - \frac{1}{2} e^{-k_1 t} \cos \left(2\pi \frac{b}{2} t \right) \right], \quad (4)$$

where k_1 and k_2 are the spin diffusion rate constants: $k_2 = 1/T_{\text{dif}}$ and $k_1 = 3k_2/2$. The ratio $k_1/k_2 = 3/2$ is sequent from the definition of the isotropic spin-diffusion

superoperator [25]. Later on, this equation was modified for the extended spin system S-I_N under MAS ([28] and references cited therein):

$$I(t) = I_0 \left\{ \frac{1}{2} \left[e^{-\frac{t}{T_{1\rho}}} - e^{-k_1 t} g_n(t) \right] + \left(\frac{S_{qe}}{\omega_{0I}} - \frac{1}{2} \right) \left[e^{-\frac{t}{T_{1\rho}}} - e^{-k_2 t} \right] \right\}, \quad (5)$$

where ω_{0I} is the Larmor frequency of I spins and S_{qe} is the quasi-equilibrium polarization of S spin that can be expressed as

$$S_{qe} = \frac{N}{N+1} \omega_{0I}. \quad (6)$$

The function $g_n(t)$ in Eq. 5 incorporates the effect of MAS on the dipolar coupling tensor components b_n ($n = 0, \pm 1, \pm 2$):

$$b_0 \Rightarrow 0; b_{\pm 1} = \frac{D_{IS}}{2\sqrt{2}} \sin(2\beta) \exp(\pm i\varphi); b_{\pm 2} = \frac{D_{IS}}{4} \sin^2 \beta \exp(\pm 2i\varphi), \quad (7)$$

where β and φ are the polar angles of $\mathbf{r}$ vector respect to MAS rotor-fixed coordinate system with z axis along the spinning axis. The exact expression of $g_n(t)$ depends on the fulfilled HH sideband matching condition $\omega_H - \omega_{IS} = n\omega_{MAS}$ [28]. The averaging of the I-S vectors can be carried out over orientations of crystallites in the case of usual powders [6, 28] or over orientations and internuclear distances for complex solids [14].

Assuming that the b -spectrum, i.e., the spectral distribution of b values and its Fourier image can be approached by the Gauss functions, the function $g_n(t)$ in Eq. 5 can be simply replaced by the Gauss decay [14]:

$$g_n(t) = e^{-\frac{t^2}{2T_2^2}}, \quad (8)$$

where the time constants T_2 is assumed to be approximately equal to the spin-spin relaxation time. Hence, Eq. 5 becomes:

$$I(t) = I_0 \left\{ \left[e^{-\frac{t}{T_{1\rho}}} - e^{-k_1 t} e^{-\frac{t^2}{2T_2^2}} \right] + \left(\frac{N-1}{N+1} \right) \left[e^{-\frac{t}{T_{1\rho}}} - e^{-k_2 t} \right] \right\} \quad (9)$$

and it can be rewritten as

$$I(t) = I_0 e^{-\frac{t}{T_{1\rho}}} \frac{N}{N+1} \left\{ 1 - \frac{N-1}{2N} e^{-(k_2-1/T_{1\rho})t} - \frac{N+1}{2N} e^{-(k_1-1/T_{1\rho})t} e^{-\frac{t^2}{2T_2^2}} \right\}. \quad (10)$$

It is easy to see that in the case of $k_1, k_2 \gg 1/T_{1\rho}$ and renormalizing $I_0 \rightarrow I_0 N/(N+1)$, Eq. 10 converts to Eq. 3, i.e., to the most often used equation of the I-I*-S model. If the quasi-equilibrium polarization of the S spin inside the spin system S-I_N is realized and thus Eq. 6 is valid, the parameter λ is related to the spin cluster size N :

$$\lambda = \frac{N-1}{2N}. \quad (11)$$

This differs from the above $\lambda = 1/(N+1)$ that was stated maybe intuitively [3, 4]. It becomes obvious that in both cases, λ cannot overstep $\lambda \leq 1/2$. It provides an additional constraint on the processing of CP kinetic curves. Namely, the results of the non-linear curve fitting of the model and experimental data have to be rejected or revised if the minimization routine leads to $\lambda > 1/2$. In this case, the possible physical reasons, why the spin quasi-equilibrium was not reached, should be carefully analyzed.

Equation 9 has the advantage, not only because it elucidates the physical sense of the parameter λ , but also because it provides the way to follow the convergence between non-classical and classical spin coupling models without any a priori discrimination, which of them should be valid for a system under study. It is quite logical to suppose that at the infinite enlargement of the spin cluster size, the non-classical I-I*-S model has to convert to the classical I-S one. In addition, indeed, in this case, $N \rightarrow \infty$ and the requirement of the high dilution of spin S among spins I is certainly fulfilled, i.e., $N_S/N_I \rightarrow 0$. Furthermore, if the I-S heteronuclear coupling is sufficiently reduced by high spin mobility, the CP kinetics should obey the classical I-S model (Eq. 1). To follow this conversion, let us first rewrite Eq. 9 inserting $N \rightarrow \infty$:

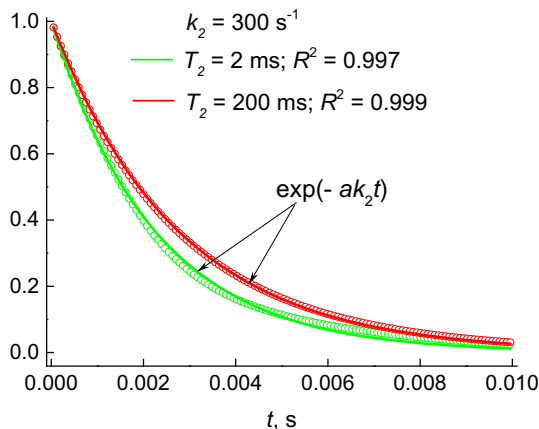
$$I(t) = 2I_0 \left\{ e^{-\frac{t}{T_{1\rho}}} - \frac{1}{2} \left(e^{-\left(k_1 t + \frac{t^2}{2T_2^2}\right)} + e^{-k_2 t} \right) \right\}. \quad (12)$$

The principal question that arises carrying out the non-linear curve fitting is: can the expression in the internal brackets (...) be recognized by the computing as the sum of two exponents or can it be merged to a single rescaled exponential decay, as

$$\frac{1}{2} \left(e^{-\left(k_1 t + \frac{t^2}{2T_2^2}\right)} + e^{-k_2 t} \right) \rightarrow e^{-ak_2 t}, \quad (13)$$

where a is the adjustable factor introduced in the sense of rescaling of the ‘true’ spin diffusion rate $k_2 = 1/T_{\text{dif}}$. The answer certainly depends on the experimental precision as well as on the parameters k_2 , $k_1 = 3k_2/2$, and T_2 . This was checked taking the set of some ‘realistic’ k_2 and T_2 , i.e., those often met in the CP kinetics experiments. The results are presented in Fig. 2. The perfect fit ($R^2 = 0.997 \div 0.999$) was obtained in the case of ‘narrow’ as well as ‘broad’ b -spectrum, i.e., short and long T_2 , varying only one parameter—the rescaling factor a (Fig. 2). This means that at the present regime of CP transfer, which is determined by k_2 and T_2 values, the two- and single- exponential decay functions (Eqs. 12 and 13) are undistinguishable, even in the case when relatively huge data sets with very high precision (computer generated curves with no random noise, 128 points per curve) are processed. Thus, it can be stated that for the large spin clusters at present conditions, the ‘classical’ and ‘non-classical’ regimes are practically

Fig. 2 Imitation of two exponential term in Eq. 13 (computer generated points shown by *open circles*) by the single exponent with the rescaled spin diffusion rate ak_2 (*solid lines*). The different T_2 values correspond to ‘broad’ and ‘narrow’ distributions of the dipolar splitting. The adjusted a values are within 1.2–1.5. More comments in text



undistinguishable. Equations 1 and 12 become equivalent, and moreover—the CP rate constants $1/T_{\text{IS}}$ and $1/T_{\text{dif}}$ used in those equations are bridged by the relation:

$$\frac{1}{T_{\text{IS}}} \rightarrow ak_2 = \frac{a}{T_{\text{dif}}}. \quad (14)$$

3.2 Processing of CP Kinetics

The experimentally measured CP MAS signals of ^{11}B in the ionic liquid [bmim][BF₄] in MCM-41 and SBA-15 were decomposed into two Lorentz-shaped ‘narrow’ and ‘broad’ components (Fig. 1). The CP kinetics was studied for each signal separately. The results are presented in Fig. 3. Indeed, both spin coupling models, discussed above, can be identically good fitted (getting almost the same R^2 values) to the experimental curves, and thus, classical and non-classical CP transfer regimes are practically undistinguishable for the studied systems. However, the non-classical model seems to be more preferable, because it directly provides T_{dif} values, i.e., the parameter originated from the exact microscopic quantum mechanical model [25], whereas T_{IS} was introduced in the more phenomenological way [24]. Despite the adjusted rescaling factor a values were found to be close to 1 (see the caption to Fig. 2) and thus $T_{\text{IS}} = T_{\text{dif}}/a \approx T_{\text{dif}}$, the non-classical treatment allows to avoid the need to elucidate the physical sense of such a rescaling. The fit parameters of the CP kinetic curves (Fig. 3), which were obtained using Eq. 9, are presented in Table 1.

The presence of two ^{11}B signals points towards the possibility of bimodal distribution of [bmim][BF₄] in mesostructured silica. Indeed, both samples can be considered as those with intermediate contents of RTIL in pores. Hence, two signals in the resolved ^{11}B CP MAS spectra could correspond to two different RTIL environments therein. Note, the presence of two different water environments was also deduced in the same silica analyzing ^1H MAS spectra [20]. The two ^{11}B signals observed in the present work can be originated from [bmim][BF₄] placed in the layer on the pore surfaces (‘broad’ component) and in the inner pore space (‘narrow’

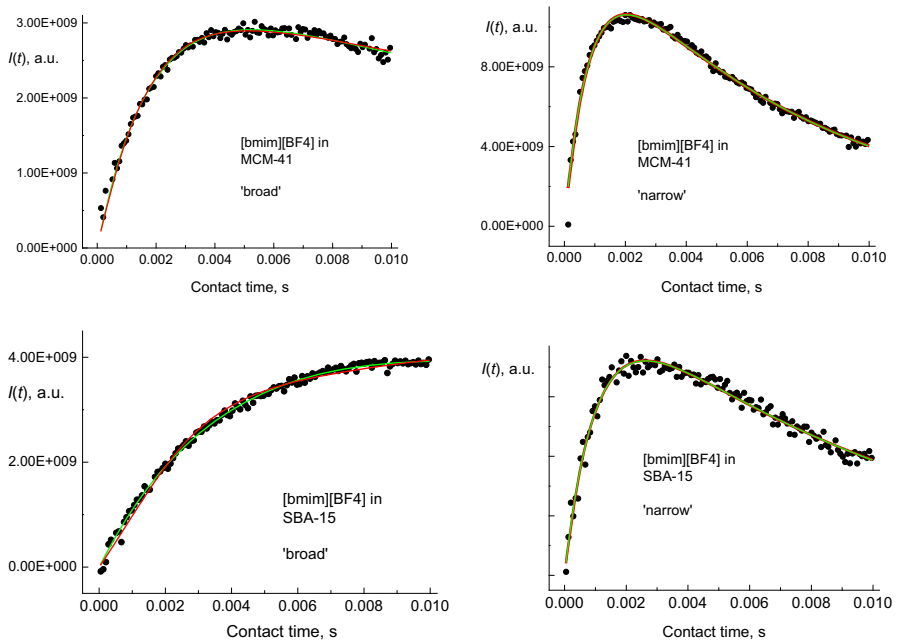


Fig. 3 $^1\text{H} \rightarrow ^{11}\text{B}$ CP kinetics upon 5 kHz MAS of [bmim][BF₄] confined in MCM-41 and SBA-15 pores. For both samples, the ‘classical’ (Eq. 1, red line) and ‘non-classical’ (Eq. 9, green line) CP transfer regimes are practically undistinguishable. The fit parameters are given in Table 1

Table 1 Fit parameters (Eq. 9) of the CP kinetic curves presented in Fig. 3

	[bmim][BF ₄] in MCM-41		[bmim][BF ₄] in SBA-15	
	Broad	Narrow	Broad	Narrow
I_0 , a. u.	$(3.66 \pm 0.07) \times 10^9$	$(1.6 \pm 0.2) \times 10^{10}$	$(4.37 \pm 0.04) \times 10^9$	$(1.05 \pm 0.02) \times 10^{10}$
N	∞	∞	∞	∞
k_2 , s ⁻¹	374 ± 15	745 ± 24	164 ± 5	617 ± 27
T_2^a , s	0.0026	0.02	0.0022	0.022
$T_{1\rho}$, s	0.031 ± 0.002	0.0072 ± 0.0001	— ^b	0.01 ± 0.003
R^2	0.984	0.988	0.992	0.971

^a T_2 values were determined from the corresponding line widths (Fig. 1) using $T_2 \approx 1/\pi\Delta\nu$

^b The value is not determined because of too short contact time range

component), respectively. It is logical to suppose that the structural organization and the mobility of [bmim][BF₄] in the inner space should be more or less similar to those in the neat RTIL and that causes the signal narrowing (Fig. 1). However, then, it is interesting to note that the spin diffusion is faster for the narrower spectra component (see the rates k_2 in Table 1), i.e., for that associated with ‘more-liquid’ fraction in the inner pore space. It is, seemingly, in contradiction to faster spin

diffusion that is expected in less mobile surface layer. This behaviour can be supposed to be a consequence of separation of RTIL anions and cations in the surface layer in contrast to ion pairing in the inner space. Comparing the kinetic data for both silica, it is seen that the spin diffusion rates (k_2) and the time constants T_2 (approx. the spin–spin relaxation time) in the inner layers in MCM-41 and SBA-15 are similar, with the differences only slightly out of the margin of error (Table 1). The main difference is seen in the spin diffusion rates in the layers on pore surfaces of MCM-41 and SBA-15. Namely, in MCM-41, it is more than twice faster than in SBA-15 (Table 1). This could be caused by the higher roughness of the silica surface of SBA-15 due to the incomplete silica layers on the inner pore surfaces [30]. The existence of these incomplete layers might explain the fact that on the pore surface of SBA-15 despite of drying, there still exist some stronger bound water molecules, i.e., those bound by two hydrogen bonds to the surface. They can be removed by the heat gun only [20]. These water molecules are tentatively assigned to surface defects that are present in SBA-15, but not in MCM-41. Hence, the difference in the spin diffusion rate and, thus, the different $^1\text{H} \rightarrow ^{11}\text{B}$ CP kinetics in [bmim][BF₄] confined in MCM-41 and SBA-15 can be caused by the additional contribution from less mobile protons on the surfaces that can act in CP transfer beside the protons in the bmim⁺ cations.

Unfortunately, no defined conclusion concerning possible differences in the mobility and relaxation mechanisms for protons in MCM-41 and SBA-15 could be done analyzing $T_{1\rho}$ values (Table 1). For the ‘narrow’ component, these values are practically the same. However, for the ‘broad’ component, this requires CP kinetics experiments which would be carried out over a much wider contact time range to be sure the plateau or the intensity decay was reached at a longer contact time.

4 Concluding Remarks

A tight convergence of ‘classical’ and ‘non-classical’ spin coupling models has been deduced for the RTIL system ([bmim][BF₄]) confined in mesoporous silica of MCM-41 and SBA-15 types. Both model dependences can be identically good fitted to the experimental curves getting almost the same correlation coefficient R^2 values and no systematic (non-random) deviations between theory and experiment appear over a wide range of contact time. However, the non-classical model seems to be more preferable, because it directly provides the information on the spin diffusion rate that is originated from the exact microscopic quantum mechanical model.

This means that the RTIL studied behaves quite differently from other well-known nano-structured systems, where the spin systems exhibit definitely non-classical spin coupling regime. Such a RTIL behaviour can be caused by several factors: (i) the requirement for high dilution of spin S among spins I is certainly fulfilled (^{11}B is ‘diluted’ among 15 protons in each bmim⁺ cation and those tentatively supposed to be on the pore surfaces); (ii) high mobility, i.e., relatively weakly hindered displacement of the BF₄[−] anion respect to the cation and dynamics of the anion along the pore surfaces deletes the borders between spin clusters, and thus, the spin cluster size $N \rightarrow \infty$.

The bimodal distribution of [bmim][BF₄] in mesostructured silica has been resolved: two different RTIL environments, viz., the layer on the pore surface and the inner pore space can be recognized. The spin diffusion along the pore surfaces in MCM-41 is more than twice faster than in SBA-15.

Finally, an interesting point is worth noting—it is well known that the geometric confinement imposed by the surface causes a strong depression of the melting point of the surface phase, e.g., the cases of water and small organic molecules in mesoporous carbon materials [31]. The studied ionic liquid system looks like an intriguing exception from this rule. Its NMR signals in the studied porous silica are more solid-like than those observed in the neat compound, i.e., in the liquid phase. These studies have to be continued enrolling a broader set of ionic liquids.

The experimental data and interpretation presented in this work should be considered only as the first step demonstrating the promising capabilities of CP MAS kinetics in the studies of such novel composites. This study will be continued by varying a level of pore filling and other experimental conditions, e.g., temperature.

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References

1. K.J.D. MacKenzie, M.E. Smith, *Multinuclear Solid-State Nuclear Magnetic Resonance of Inorganic Materials* (Pergamon Press, New York, 2002)
2. E.O. Stejskal, J.D. Memory, *High Resolution NMR in the Solid State. Fundamentals of CP/MAS* (Oxford University Press, Oxford, 1994)
3. W. Kolodziejski, J. Klinowski, *Chem. Rev.* **102**, 613 (2002)
4. J. Kolmas, W. Kolodziejski, *Chem. Phys. Lett.* **554**, 128 (2012)
5. G. Mali, V. Kaučič, *J. Chem. Phys.* **117**, 3327 (2002)
6. G. Mali, N. Rajić, N. Zabukovec Logar, V. Kaučič, *J. Phys. Chem. B* **107**, 1286 (2003)
7. A. Kafilak, W. Kolodziejski, *Magn. Reson. Chem.* **46**, 335 (2008)
8. M. Hologne, P. Bertani, T. Azraïs, C. Bonhomme, J. Hirschinger, *Solid State Nucl. Magn. Reson.* **28**, 50 (2005)
9. C. Jäger, T. Welzel, W. Meyer-Zaika, M. Eppe, *Magn. Reson. Chem.* **44**, 573 (2006)
10. A. Vyalikh, P. Simon, T. Kollmann, R. Kniep, U. Scheler, *J. Phys. Chem. C* **115**, 1513 (2011)
11. A. Vyalikh, P. Simon, E. Rosseeva, J. Buder, R. Kniep, U. Scheler, *J. Phys. Chem. B* **118**, 724 (2014)
12. S. Kareiva, V. Klimavicius, A. Momot, J. Kausteklis, A. Prichodko, L. Dagys, F. Ivanauskas, S. Sakirzanovas, V. Balevicius, A. Kareiva, *J. Mol. Struct.* **1119**, 1 (2016)
13. V. Klimavicius, L. Dagys, V. Balevicius, *J. Phys. Chem. C* **120**, 3542 (2016)
14. L. Dagys, V. Klimavicius, V. Balevicius, *J. Chem. Phys.* **145**, 114202 (2016)
15. ILSEPT, in *3rd International Conference on Ionic Liquids in Separation and Purification Technology*, Kuala Lumpur, Malaysia, 8–11 Jan 2017. <http://www.ilsept.com/conference-program.html>
16. P. Wasserscheid, T. Welton (eds.), *Ionic Liquids in Synthesis*, 2nd edn. (Wiley, Weinheim, 2008)
17. H. Weingärtner, *Angew. Chem. Int. Ed.* **47**, 654 (2008)
18. D. Bankmann, R. Giernoth, *Progr. Magn. Reson. Spectrosc.* **51**, 63 (2007)
19. M.A. Firestone, P. Thiyagarajan, D.M. Tiede, *Langmuir* **14**, 4688 (1998)

20. B. Grünberg, T. Emmeler, E. Gedat, I. Shenderovich, G.H. Findenegg, H.H. Limbach, G. Buntkowsky, *Chem. Eur. J.* **10**, 5689 (2004)
21. E. Gedat, A. Schreiber, G. Findenegg, I. Shenderovich, H.H. Limbach, G. Buntkowsky, *Magn. Reson. Chem.* **39**, 149 (2001)
22. J. Gao, N.J. Wagner, *Langmuir* **32**, 5078 (2016)
23. W. Liu, L. Cheng, Y. Zhang, H. Wang, M. Yu, *J. Mol. Liq.* **140**, 68 (2008)
24. S. Ando, R.K. Harris, S.A. Reinsberg, *J. Magn. Reson.* **141**, 91 (1999)
25. L. Müller, A. Kumar, T. Baumann, R.R. Ernst, *Phys. Rev. Lett.* **32**, 1402 (1974)
26. R.K. Hester, J.L. Ackermann, V.R. Cross, J.S. Waugh, *Phys. Rev. Lett.* **34**, 993 (1975)
27. A. Naito, C.A. Mc Dowell, *J. Chem. Phys.* **84**, 4181 (1986)
28. C.A. Fyfe, A.R. Lewis, J.M. Chézeau, *Can. J. Chem.* **1999**, 77 (1984)
29. V. Klimavicius, A. Kareiva, V. Balevicius, *J. Phys. Chem. C* **118**, 28914 (2014)
30. I. Shenderovich, G. Buntkowsky, A. Schreiber, E. Gedat, S. Sharif, J. Albrecht, N.S. Golubev, G.H. Findenegg, H.H. Limbach, *J. Phys. Chem. B* **107**, 11924 (2003)
31. Y. Xu, T. Watermann, H.H. Limbach, T. Gutmann, D. Sebastiani, G. Buntkowsky, *Phys. Chem. Chem. Phys.* **16**, 9327 (2014)